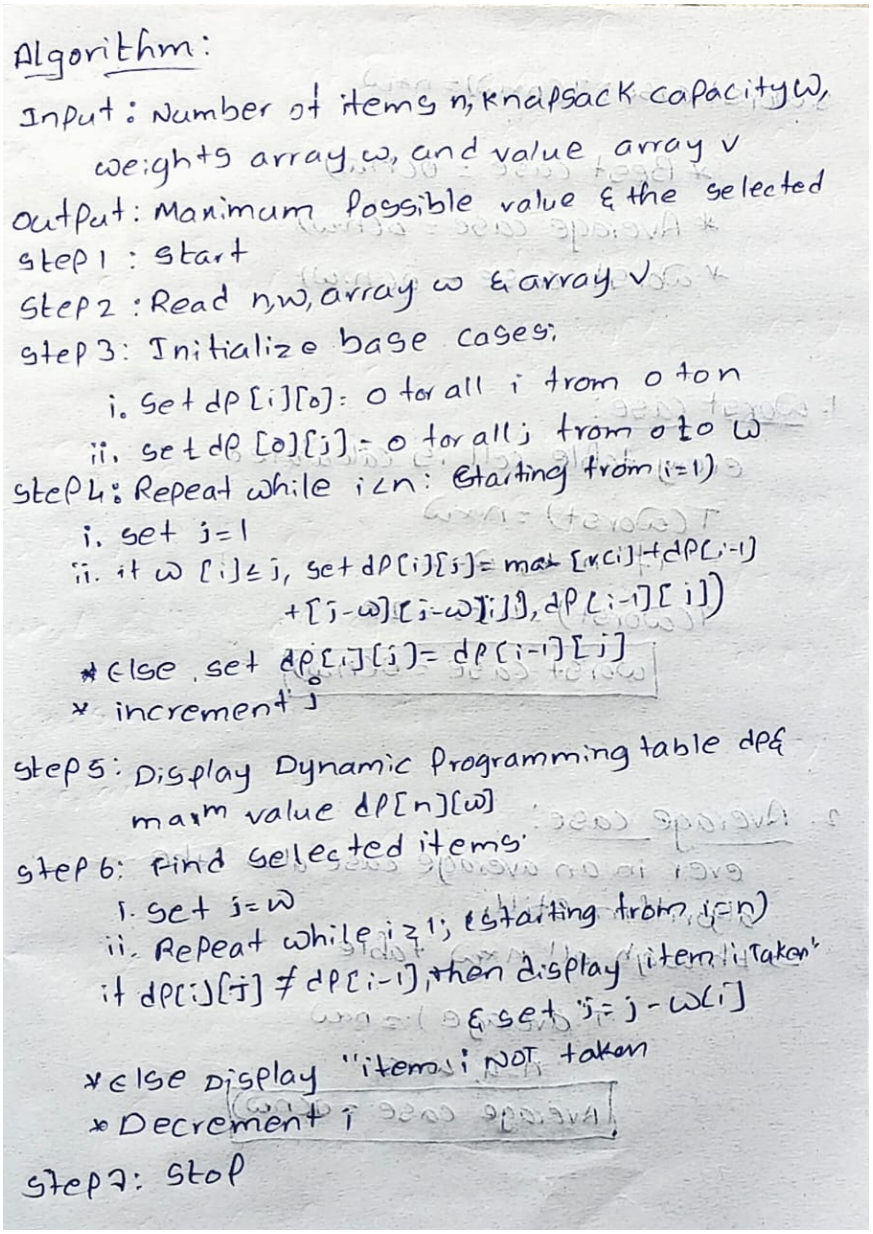
PRACTICAL-5

Aim : To find the maximum value of items that can be placed in a knapsack of capacity **W** without exceeding its weight limits, and to identify which items are selected, using Dynamic Programming.

Algorithm :



Algorithm:
Input: Number of items n, knapsack capacity W, weights array w, and value array v
Output: Maximum possible value & the selected
step 1 : start
Step 2 : Read n, W, array w & array v
step 3: Initialize base cases:
i. Set dp[i][0] = 0 for all i from 0 to n
ii. set dp[0][j] = 0 for all j from 0 to W
Step 4: Repeat while i<n: (starting from i=1)
i. set j=1
ii. if w[i]≤j, set dp[i][j] = max[v[i]+dp[i-1]
+[j-w][j-w][i]], dp[i-1][i])
* else set dp[i][j] = dp[i-1][j]
* increment j
step 5: Display Dynamic Programming table dpf
max value dp[n][w]
step 6: Find selected items:
i. set j=w
ii. Repeat while i≥1; (starting from i=n)
if dp[i][j] ≠ dp[i-1], then display "item i taken"
& set j = j - w[i]
* else display "item i NOT taken
* Decrement i
Step 7: Stop

Program / Code :

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

class Knapsack {

private:
    vector<int> w;            // weights
    vector<int> v;            // values
    int n, W;                 // number of items and capacity
    vector<vector<int>> dp;   // DP table

public:

    // Get Input
    void getInput() {

        cout << "Enter number of items : ";
        cin >> n;

        cout << "Enter knapsack capacity : ";
        cin >> W;

        w.resize(n + 1);
        v.resize(n + 1);

        // Creating DP table
        // Time Complexity = O(nW)
        dp.assign(n + 1, vector<int>(W + 1, 0));

        for (int i = 1; i <= n; i++) {
            cout << "Item " << i
                 << " weight and value : ";
            cin >> w[i] >> v[i];
        }
    }

    // Solve Knapsack using Dynamic Programming
    void solve() {

        // Time Complexity = O(nW)
        for (int i = 1; i <= n; i++) {
```

```
        for (int j = 1; j <= W; j++) {

            if (w[i] <= j) {

                dp[i][j] = max(
                    v[i] + dp[i - 1][j - w[i]],
                    dp[i - 1][j]
                );

            } else {

                dp[i][j] = dp[i - 1][j];
            }
        }
    }
}

// Print DP Table
void printTable() {

    cout << "\nDP Table:\n ";

    for (int j = 0; j <= W; j++) {
        cout << j << " ";
    }

    cout << endl;

    // Time Complexity = O(nW)
    for (int i = 0; i <= n; i++) {

        cout << "i=" << i << " ";

        for (int j = 0; j <= W; j++) {
            cout << dp[i][j] << " ";
        }

        cout << endl;
    }
}

// Find Selected Items
void findItems() {
```

```
cout << "\nSelected Items:\n";

        int j = W;

        // Time Complexity = O(n)
        for (int i = n; i >= 1; i--) {

            if (dp[i][j] != dp[i - 1][j]) {

                cout << "Item " << i
                    << " (w=" << w[i]
                    << ", v=" << v[i]
                    << ") TAKEN\n";

                j -= w[i];

            } else {

                cout << "Item " << i
                    << " (w=" << w[i]
                    << ", v=" << v[i]
                    << ") NOT taken\n";
            }
        }
    }

    // Show Result
    void showResult() {

        cout << "\nMaximum Value = "
            << dp[n][W] << endl;
    }
};

int main() {

    Knapsack k;

    k.getInput();
    k.solve();
    k.printTable();
    k.showResult();
    k.findItems();
```

```
    return 0;
}
```

Output :

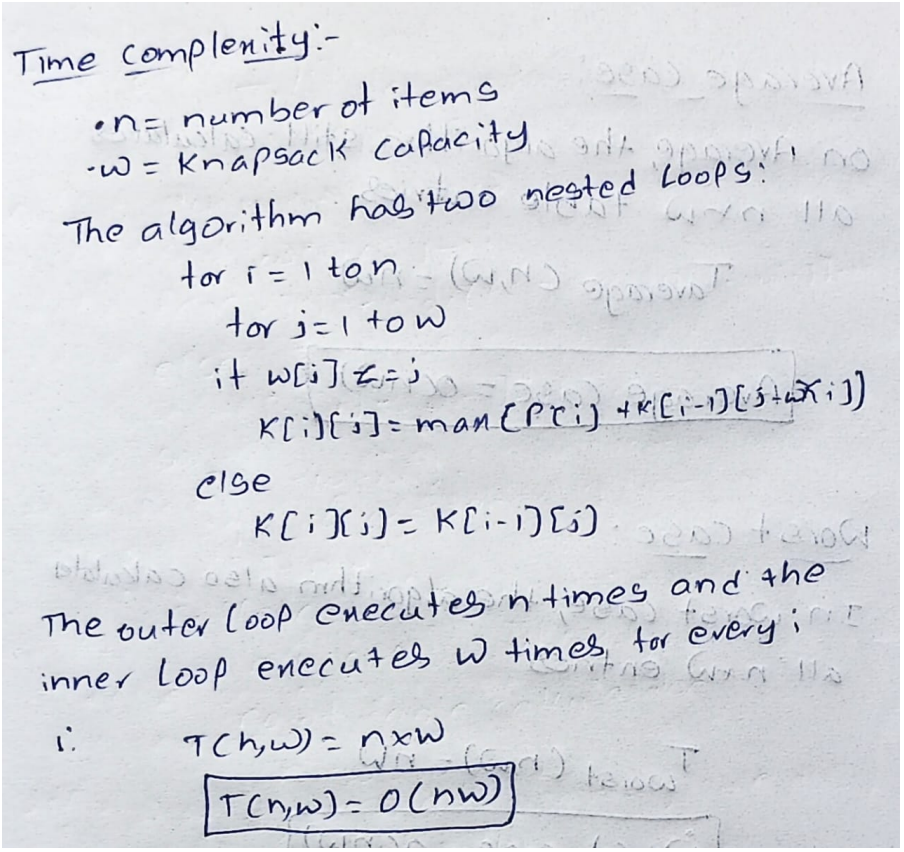
```
Enter number of items : 3
Enter knapsack capacity : 5
Item 1 weight and value : 2
1
Item 2 weight and value : 6
3
Item 3 weight and value : 1
4

DP Table:
 0 1 2 3 4 5
i=0 0 0 0 0 0 0
i=1 0 0 1 1 1 1
i=2 0 0 1 1 1 1
i=3 0 4 4 5 5 5

Maximum Value = 5

Selected Items:
Item 3 (w=1, v=4) TAKEN
Item 2 (w=6, v=3) NOT taken
Item 1 (w=2, v=1) TAKEN
```

Time Complexity :



Time complexity:-

- n = number of items
- w = knapsack capacity

The algorithm has two nested loops:

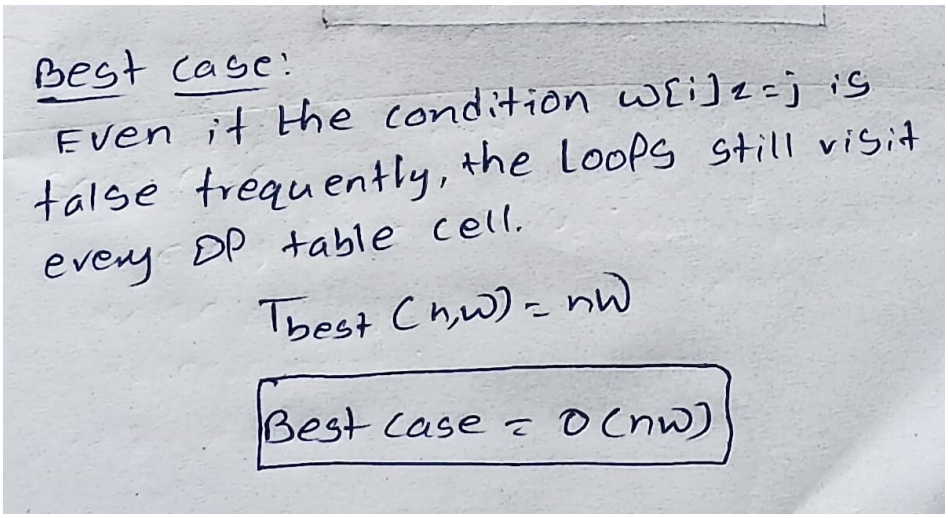
```
for i = 1 to n
  for j = 1 to W
  if w[i] <= j
    K[i][j] = max(P[i] + K[i-1][j-w[i]])
  else
    K[i][j] = K[i-1][j]
```

The outer loop executes n times and the inner loop executes W times for every i.

i. $T(n,w) = n \times w$

$T(n,w) = O(nw)$

1.BEST CASE:



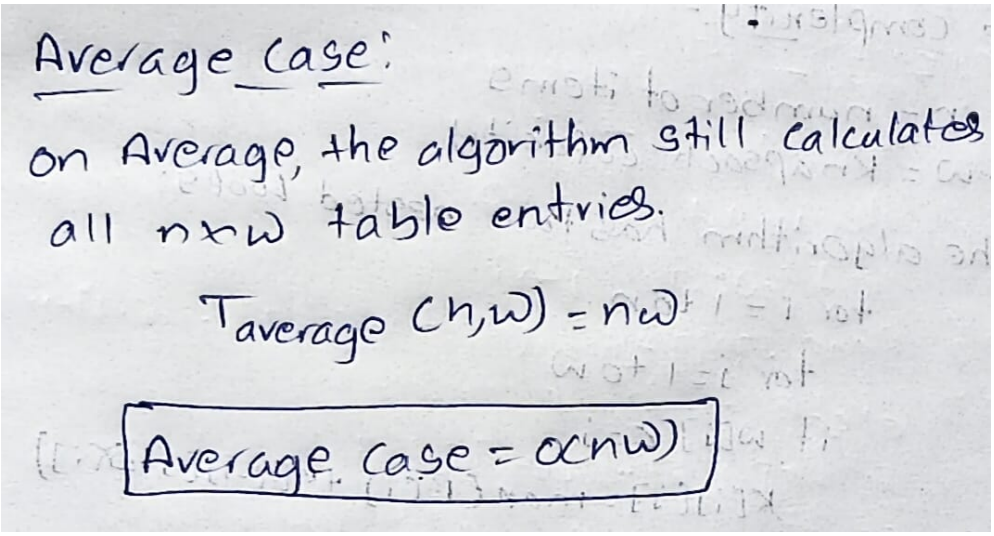
Best case:

Even if the condition w[i] <= j is false frequently, the loops still visit every DP table cell.

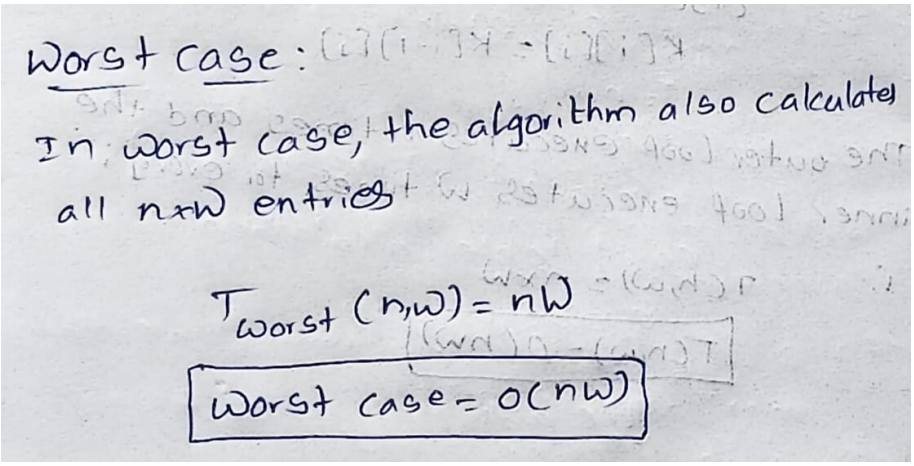
$T_{best}(n,w) = nW$

Best case = $O(nw)$

2.AVERAGE CASE:



3.WORST CASE:



Conclusion :

The 0/1 Knapsack problem was successfully solved using Dynamic Programming. The algorithm finds the maximum possible profit by systematically filling a DP table. It has a fixed time complexity of O(n × W) for all cases, since the program evaluates each entry of the table to determine the best possible solution.